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REVIEW

Overview of frequency control techniques in power systems with high inverter-based resources: Challenges and mitigation measures

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Abstract

Power systems are rapidly transitioning towards having an increasing proportion of electricity from inverter-based resources (IBR) such as wind and solar. An inevitable consequence of a power system transition towards 100% IBR is the loss of synchronous generators with their associated inertia, frequency, and voltage control mechanisms. To ensure future grid needs can be met, the source of system services to meet these needs must change from synchronous plants to IBR. Under this context, the main objective is to extensively review grid frequency stability challenges concerning the massive integration of IBR from the perspective of system operators. Following that, the newly established international fast frequency response services in different renewable dominant power systems to address low inertia challenges are compared from various perspectives. Finally, potential solutions based on various renewable enabling technologies and renewable sources are investigated from the standpoint of their expectation to play an active role in the energy system and to provide a wide range of system frequency services. The findings of this study provide a deep understanding of prospective frequency control solutions and an actionable dataset that will support various research and policy needs as we approach net-zero targets.

KEYWORDS

energy storage, frequency control, frequency response, power system transient stability, renewable energy sources

1 | INTRODUCTION

The global energy sector must be fundamentally restructured to achieve complete decarbonisation, as outlined in the Paris Agreement of December 2015. To limit global temperature rise to less than 2 degrees Celsius (°C), with a target of below 1.5° C, participants committed to taking necessary actions [1]. The Intergovernmental Panel on Climate Change 2018 assessment was clear that the world must be carbon-neutral by 2055 to endure Paris Agreement and minimise the potential impacts of catastrophic global warming. [2]. Renewable energy has been considered a critical driver in meeting these objectives, particularly for electrical power systems, which are among the easiest energy supplies to decarbonise. Global installed renewable electricity, including renewable generation and its share of overall power generation, was 2802 GW, 7643 TWh, and 28.6% in 2020, as depicted in Figure 1 [3, 4].

The Paris Agreement aims to increase worldwide annual renewable electricity generation from 28% today to 86% by 2050, with distributed generation (DG) based on variable renewable (i.e. wind and solar photovoltaic (PV)) projected to account for about 70% and about 60%, respectively [5]. Regional efforts are also worth mentioning. The European Union (EU) countries have set ambitious targets to produce 54% of the gross generated electricity from renewable sources by 2030 [6] and reach net zero by 2050. Distributed generation-based wind power and PV systems are key drivers towards these objectives accounting for 56% of Europe's electricity by

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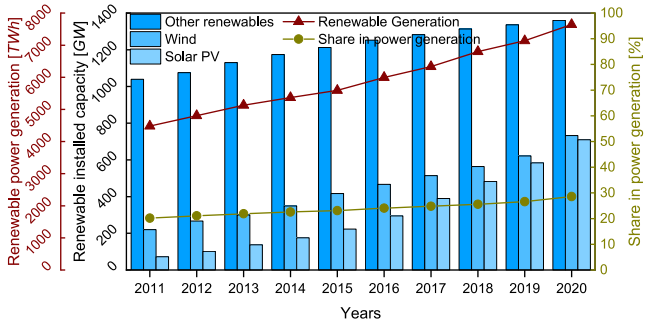


FIGURE 1 Global renewable installed capacity, generation, and its share in the final electricity generation.

2030 and increased upto 27% from 2021 [7]. Among the EU countries, renewable generation in Ireland accounted for 43% of the overall electricity consumption in 2020 [8]. Ireland has recently released the Climate Action Plan 2021 (CAP21), which sets itself the challenge of sourcing 70%, which may increase to 80%, of its electricity generation from renewable energy, predominantly through variable speed wind turbines (VSWT), by 2030 [9]. The CAP21 also contains a higher commitment to deploy up to 5 GW of offshore wind installed capacity. The United Kingdom (UK) government aims to achieve net zero carbon emissions by 2050, with no precise target set for Northern Ireland, which may reach 80% by 2030 [10].

As power systems shift to high levels of variable DG penetration, the technical characteristics of the network are changing, posing grid instability and vulnerability challenges due to the intermittency and non-synchronous nature of renewable energy if operated in the grid-following mode (GFL) [11]. Uncertainties associated with the operation of variable DG systems are currently handled by utilising the spinning reserve of conventional generators [12]. However, significant penetrations of DG systems require predictability to prevent the possibility of a mass blackout. The challenge is managing grid reliability with the non-synchronous nature of DG systems and high voltage direct current (HVDC) interconnectors, which are mostly inverter-based resources (IBR) that operate in GFL mode. This displaces the amount of spinning mass and reduces the overall power system inertia, making the system more reluctant to a change in angular velocity and frequency [13]. This may require renewable generations to be curtailed, forcing transmission system operators (TSO) to rely on more fossil fuel-based plants until adequate inertia and reserves are available to manage contingency events and keep the system running at a nominal frequency.

Variable renewable IBR replacing synchronous generators lead to a loss of both synchronous inertia response (SIR) and primary operating reserve (POR) capabilities, making it challenging for TSOs to maintain stability during severe contingencies. Low system inertia and spinning reserve increase the rate of change of frequency (RoCoF) and frequency deviation (nadir), which can trigger anti-islanding DG protection relays, leading to load shedding and eventually complete system blackout [14, 15]. To address these issues, a fast frequency response (FFR) ancillary services market has been established in some island countries to help TSOs maintain a stable operating frequency and facilitate the integration of high IBR [16].

In addition to transmission system changes, the electrification of end-use sectors, such as transportation (i.e. electric vehicles), building (i.e. heat pumps, and industry (i.e. data centres) is changing the distribution network and creating a need for system flexibility. This flexibility will come from various sources and be used for transmission and distribution system challenges. The International Renewable Energy Agency (IRENA) defines flexibility as the capability of a power system to cope with the variability and uncertainty that solar and wind energy introduces at different time scales while minimising renewable power curtailment and reliably delivering energy demand to all customers [17]. However, reaching this goal will require load centres on the grid to be more flexible for three reasons, (1) better match the grid's increasingly unpredictable supply, (2) fine-tune the grid for transient stability, (3) and control power flows on the network architecture. These requirements will not only become more pronounced in power systems with high IBR, but the conventional fossil-fuel generators that will be employed to address these flexibility needs will also become more expensive to operate. As a result, other resources (i.e. synchronous condensers, energy storage, and demand response) will be required to provide power system frequency needs by participating in the ancillary services [18, 19].

The majority of the issues, including the effect of IBR on power system frequency stability [20], inertia level [21], and voltage control mechanisms [22], have been addressed in recent research. However, none of them evaluated the challenges and potential technology solutions in the context of TSOs' plans to achieve net-zero targets. Different from the literature, the main objective of this study is to comprehensively review strategies used to improve frequency control and manage RoCoF in both islanded and interconnected power systems. The scope of the analysis then extends to frequency support mechanisms from wind and solar systems, HVDC interconnectors, synchronous and asynchronous condensers, and battery energy storage systems (BESS) equipped with both grid forming (GFM) and GFL converters.

The remainder of this study is structured as follows. Section 2 examines the impact of IBR on frequency stability and RoCoF management in weak and well-interconnected power systems. In Section 3 deep understanding of international frequency service and frequency response enabling technologies are critically evaluated. Discussions and recommendations are provided in Section 4. Section 5 concludes the paper with some remarkable future work directions.

2 | IMPACTS OF INVERTER-BASED RESOURCES ON FREQUENCY STABILITY

The state-of-the-art DG-based renewable generation technologies (i.e. wind and solar PV plants) are integrated into the grid through power converters. The massive integration of these DG systems into the transmission and distribution networks

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impacts the overall dynamic stability of the power system [23]. This is primarily caused by two main changes in the network topology: displacement of conventional power plants and rapid time constants introduced by power electronic converters. Furthermore, power systems are shifting from centralised massive synchronous generators to decentralised generation at the distribution level [24]. The latter poses a two-fold challenge: (1) observing the generation fleet in medium and low voltage network infrastructure becomes difficult for TSOs, and (2) load flow patterns in transmission and distribution networks become volatile, leading to greater reverse flows and faster voltage instability across the network nodes [25]. All these changes introduce several stability issues that are commonly classified, as Figure 2 illustrates [26].

With the high penetration of IBR, two new components of power system stability have recently emerged: (1) converter-driven stability (oscillations due to cross-coupling between dynamic electromechanical devices and network transient), and (2) resonance stability (oscillations due to periodic and inadequate system energy dissipation) [26]. Finally, short term frequency stability is explained in the following sections, while long term frequency stability, voltage and rotor angle stabilities are out of the scope of this paper. Grid frequency is crucial for power systems as it determines the balance between generation and demand and should be kept within a permissible range. High power imbalances lead to larger frequency deviations, with under frequency being a challenge, especially during high intermittent renewable penetrations [27]. Over frequency is easier to manage since generation technology can swiftly decrease output power and rebalance supply and demand [28].

2.1 | Frequency response of conventional power systems

Frequency deviations are caused by disturbances and must be addressed by ancillary system services at the transmission level. Synchronous generators' POR is traditionally used to contain frequency deviations during the arresting period, with secondary and tertiary responses activated to recover frequency back to a quasi-steady state [29]. If the primary response fails, under frequency load shedding (UFLS) may be triggered, potentially resulting in further cascade issues similar to the disruption seen in the UK in 2019 [30]. To protect against such events, the TSOs of Ireland and Northern Ireland established a comprehensive work plan called Delivering a Secure, Sustainable Electricity System (DS3) in 2011 [31], to tackle power quality issues and securely operate at system nonsynchronous penetration (SNSP) up to 75%. The TSOs have designed and implemented new frequency control services to address a wide range of applications and timelines.

An example of the DS3 system services in the Irish power system and detailed control responses that are used during each time scale is depicted in Figure 3. The graph illustrates a typical frequency response following a large generation loss, resulting in frequency deviation from the nominal value. During the arresting phase up to the nadir, the SIR and FFR services act to slow the RoCoF. Although SIR is useful, it alone cannot prevent frequency deviation, and other measures are necessary to act during this period. Following the SIR service, thermal power plants with turbine governors respond to system frequency during this phase, providing POR to arrest frequency deviation. The POR is a contingency reserve responsible for arresting the initial drop and restoring the frequency, which occurs at the level of individual controllers via droop control mechanisms.

At frequency nadir, a new rebalance between generation and demand typically occurs. Various factors affect the severity of nadir and none of them can be examined in isolation without the other being considered. These factors include the overall system inertia, disturbance size, response speed of POR (i.e. gas or hydro turbines), the magnitude of response, and load sensitivity to frequency deviation [32]. A combined response of POR and secondary operating reserve (i.e. automatic generation control)

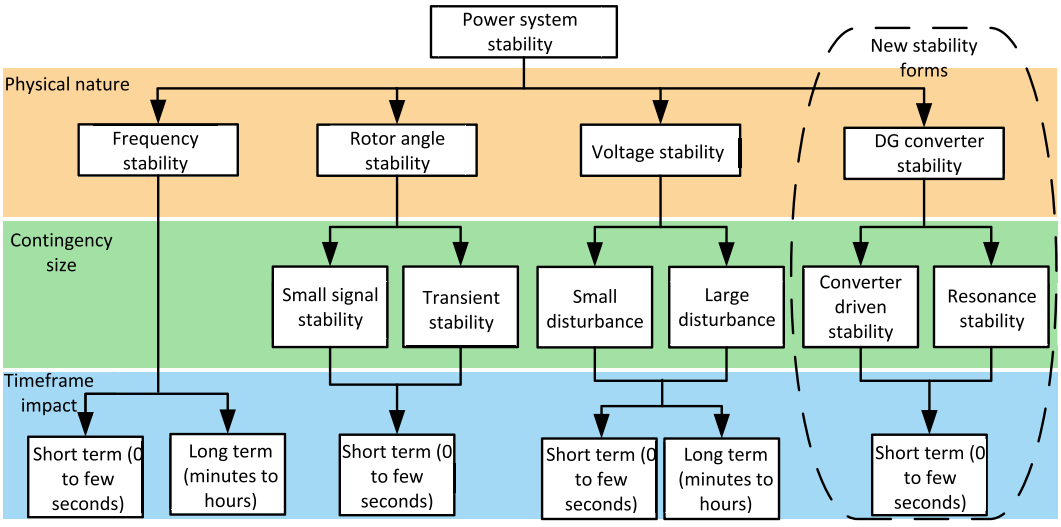


FIGURE 2 Classification of power system stability issues.

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in the timescale 5–15 s and 15–90 s, respectively, eventually recover frequency to a post-disturbance value.

Traditional thermal power plants provide many of these services, but their prevalence reduces the system's overall inertia, resulting in high RoCoF values during disturbances [33]. This means that the remaining synchronous generators will have less time to act during the arresting phase. Thus, both TSOs on the island of Ireland have established the complementary FFR service, which relies on alternative sources to provide a faster form of system reserve to bridge the gap between SIR and POR services. The FFR service is the most lucrative, requiring active power to be injected in less than 2 s from the time of the disturbance in response to both over and under frequency events [34]. The FFR service overlaps with SIR and POR, and its effectiveness depends on the amount and quality of POR supplied, with faster POR minimising required FFR capacity at any given inertia level [35].

2.2 | Power system inertia

Inertia is an inherent characteristic of spinning machines in a power system that resists changes in rotational speed [36]. It is the physical behaviour of the system that plays an essential role in limiting RoCoF during disturbances. Inertia is available immediately and is typically provided by synchronous generators, synchronous condensers, and some spinning demand loads. It is a key determinant of the strength and stability of the power system. The amount of inertial response in a system is proportional to the total number of synchronous generators, turbines, and the speed of rotation of each rotating mass, which can be determined using Equation (1).

$$H_{sys} = \frac{E_k}{S_{sys}} = \frac{1}{2}\frac{\omega^2}{S_{sys}} \sum_{i=1}^{n} J_i \quad \forall\, i \in R \tag{1}$$

where H_{sys} is the system inertia constant in (s), E_k denotes the total kinetic energy in (Joule), S_{sys} is the system base in (MVA), ω presents the rotor rotational speed in (rad/s), and J_i moment of inertia of synchronous machine i in (kg.m^2), n is the number of online spinning machines. From Equation (1), the complete system inertia constant H_{sys} containing n numbers of machines can be reformulated using Equation (2).

$$H_{sys} = \sum_{i=1}^{n} H_i.S_i/S_{sys} \tag{2}$$

where H_i inertia constant of machine i, S_i nominal power of machine i. The level of H_{sys} is an indicator that defines the degree and quantifies robustness of system frequency in interconnected networks. The RoCoF in the frequency arresting phase is a direct metric that reflects the amount of available H_{sys} as in Equation (3).

$$\frac{df}{dt}\Big|_{t=0^+} = \frac{f_o}{2H_{sys}S_{sys}}\left(P_g - P_l\right) = \frac{\Delta P_{imbalance}\, f_o}{2H_{sys}S_B} \tag{3}$$

where df/dt is the RoCoF in (Hz/s), $P_g - P_l = \Delta P_{imbalance}$ depicts the generation and demand imbalance in (MW), and f_o is the nominal frequency of the system in (Hz). Extreme RoCoF values are caused by a higher power loss or lower system inertia. The amount of H_{sys} decreases as more conventional synchronous generators are replaced by GFL-based IBR.

As seen in Figure 4, increasing the penetration of GFL-based IBR in the system results in extreme frequency deviation and RoCoF for the same disturbance [14, 15]. This is an example of a power system in which governor-equipped synchronous generators are replaced with GFL-based IBR operating at maximum power point tracking (MPPT) while the system still has a large amount of slow secondary reserve. This graph indicates that lowering inertia makes the power system faster, necessitating faster detection and less latency in RoCoF

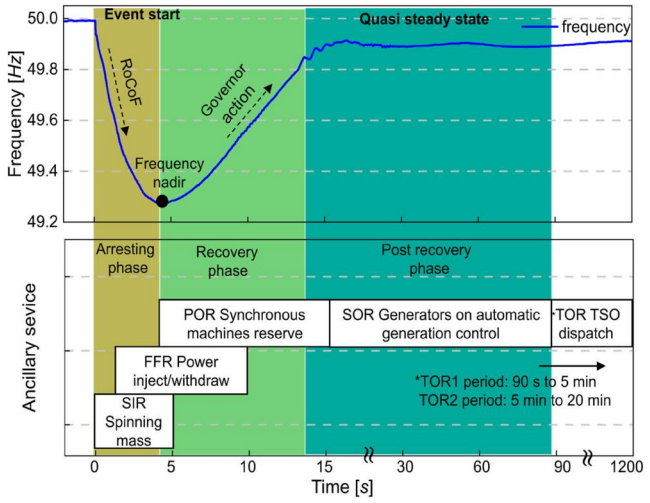


FIGURE 3 Timescale of frequency response services in the Irish power system following a typical disturbance.

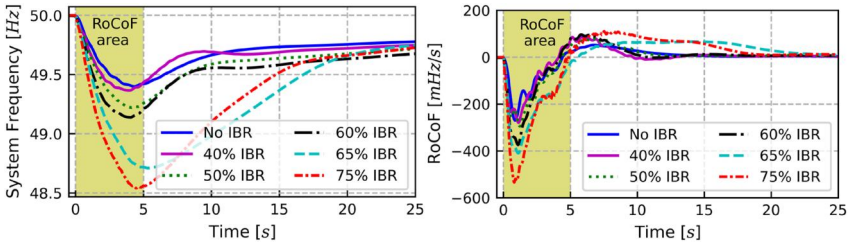


FIGURE 4 Effect of inertia and primary operating reserve (POR) reduction on the system frequency and rate of change of frequency (RoCoF).

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measurement techniques. This raises the risk of maximum RoCoF detection inaccuracy in the future [37] and a challenge for regional RoCoF variations [38]. This is because the maximum RoCoF occurs within the first second following the disturbance, and when the frequency settles at the nadir, the RoCoF becomes zero. Furthermore, a high RoCoF is typically problematic since it shortens the time window for POR service to control the frequency deviation. Physically, DG and conventional power plants are not designed to tolerate high RoCoF events.

In fact, defining the level of inertia in a power system is subjective since it depends on various network conditions, including network interconnection, system size, largest infeed units, generation/load loss, and proportion of IBR. The conventional method for determining inertia in real-time is to combine the inertia from synchronous machines. A quantified matrix is proposed by Ref. [21] to determine current and future inertia levels and identify many approaches currently employed by TSOs. This is because as part of the absolute change in system frequency, the TSO needs to manage the RoCoF variable.

2.3 | Rate of change of frequency management

2.3.1 | Rate of change of frequency in large synchronous areas

In 2013, the Nordic countries of the EU launched the 'Future System Inertia' project to 1) develop a methodology to evaluate system inertia and frequency excursions; 2) develop a real-time inertia estimator; 3) harmonise inertia calculation; 4) investigate the impact of future generation and demand changes on system inertia [39]. The project also found that the Nordic power system's inertia is projected to remain between 124 and 305 GWs, higher than the historical minimum inertia of 115 GW.s in 2009, indicating a low likelihood of severe RoCoF values [40].

Following a typical generation loss, small synchronous areas would observe rapid and substantial frequency excursions, yet large synchronous areas would not see the same size of frequency deviations unless a significant disruption occurs, such as a system split. Currently, inertia levels are adequate for the normal interconnected operation of the EU power system while considering the normative loss of 3 GW [41]. In normal interconnected operations, RoCoF and frequency deviations are not a concern since the imbalances that occur are small in comparison to the present network inertia. In the EU, a loss of 3 GW credible disturbance would result in a 0.8 Hz frequency decline, considering the self-regulating effect of the load is 1%/Hz [42]. The network will not experience an extreme frequency drop unless a significant non-credible situation occurs, such as a system split during high power import/export conditions. However, an analysis showed that a reduction in the EU system inertia is noticeable which has lowered from 92% of all generations in 2012 to 79% in 2021 and is anticipated to decrease to 44% by 2030 [7].

Among the EU countries, Germany is a leading jurisdiction in the level of renewable power integration into its network, yet no challenges are observed. There are several reasons why Germany has been able to successfully integrate significant amounts of renewable energy, including the strength of its power networks, the flexibility of coal and nuclear plants, and well synchronous interconnection with neighbouring countries [39]. This results in the overall non-synchronous penetration remaining relatively low. Thus, the system is unlikely to exhibit issues related to RoCoF until a considerably high penetration of IBR is integrated. However, to ensure dynamic stability in such circumstances, the mid-term solution may be to constrain cross-border flows between small and large synchronous regions [43]. Denmark is another example that regularly reaches or exceeds 100% non-synchronous penetration of load, with synchronous condensers used to increase system rotational inertia. Due to the significant share of IBR in the Danish system, new power plants larger than 1.5 MW must withstand RoCoF values of 2.5 Hz/s for 80 ms delay, which is a little increase over the previous 2 Hz/s limit [44].

Finally, due to IBR, North American Electric Reliability Corporation and Hydro-Québec TransÉnergie (HQT) [45] observed a decrease in their respective system inertia and frequency response, which both proposed the use of synthetic inertia as a viable remedy. The Electric Reliability Council of Texas (ERCOT), in the United States of America (USA), has recently faced some technical issues due to the high IBR [46]. Issues arose as overall ERCOT inertia declined from 152 GW.s to 130 GW.s between 2015 and 2017, which was partially attributed to the reduction in the net load [47]. This system is currently evaluating potential solutions to enhance frequency responsiveness, including renewables, and estimating prospects to secure it.

2.3.2 | Rate of change of frequency in islanded power systems

In islanded power systems that are integrated with other large power systems via HVDC interconnectors, the impact of inertia reduction is especially significant. For example, the power system of Ireland, Great Britain (GB), and South Australia have already begun to have potential issues and are becoming more vulnerable to experiencing high frequency excursions and even partial blackouts during power outages [48]. In the Irish power and GB, RoCoF limits are determined by protection settings (i.e. loss of mains (LoM)) on the DG systems [37]. This protection system disconnects DG systems when the detected RoCoF breaches the predefined RoCoF and time delay thresholds.

In GB, the LoM was triggered in 2012 due to a high RoCoF following a loss of generation associated, which required market intervention to replace non-synchronous renewable sources with high inertia thermal power plants [24]. Another example is 9 August 2019, cascading system failures in GB caused a vector shift and triggered the RoCoF protective relay, resulting in the disconnection of 350 MW of

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DG systems [30]. National Grid is currently exploring measures to address declining inertia levels, the cost of operating the system to prevent LoM protection has increased from £60 million in 2018 to £150 million in 2019 [49]. Yet, it is unclear if National Grid would approach the issue by incentivising inertia provision from synchronous sources or relying on FFR services in the future.

The problem of low inertia does not have a simple or single solution; rather, all possible options must be studied and weighed, whether now or in the future. Three alternative options are currently being considered by some TSOs [50], which are:

(1) Limiting inertia floor during normal and transient conditions.
(2) Modifying system equipment and grid codes to tolerate higher RoCoF values, as well as reconfiguring the current RoCoF settings to new standards.
(3) Incentivising new forms of fast active power to preserve the LoM through the use of ancillary services.

The first option has been implemented in Ireland, which involves maintaining the required inertia level, keeping some online synchronous generators, and limiting the RoCoF by constraining the output of the largest generating unit. For instance, the TSOs of the all-island of Ireland presently maintain at least eight large synchronous sets online at all times, which requires containing generation from renewable power to allow synchronous machines to provide rotational inertia into the balancing mix [51]. To account for this and ensure secure and prudent operation, an operational metric known as SNSP was defined in both jurisdictions, as in Equation (4) [31].

$$SNSP = \frac{(Non-synchronous\ generation + Net\ interconnector\ imports)}{(Demand + Net\ interconnector\ export)} \times 100 \quad (4)$$

The SNSP metric is a proxy for ensuring enough system services and avoiding system inertia drops below 23 GW.s [51]. The current SNSP limit is 65%, but to achieve the 70% renewable electricity targets by 2030, a 95% instantaneous SNSP is expected [52]. The main challenge for GB and Ireland is renewable curtailment caused by reduced inertia due to the high wind generation, which requires TSOs to limit renewable output power generation according to system operation rules such as the SNSP limit.

The second option is also being deployed by TSOs of some island power systems, as illustrated in Table 1. Historically, the RoCoF threshold in Ireland was −0.5 Hz/s over 500 ms [53], whilst, in GB, it was −0.125 Hz/s with no time delay [30]. This was in the context of systems with high inertia and many synchronous generators. However, with diminishing inertia, operating the system under the −0.125 Hz/s threshold has become increasingly difficult. Therefore, the RoCoF parameters in Ireland and GB have recently been relaxed to 1 Hz/s 500 ms to reduce ancillary service costs [49]. This may temporarily alleviate the RoCoF security issue, but it is not a silver bullet as several countries have already planned RoCoFs above 1 Hz/s [24]. This is because a further reduction in the RoCoF setting may jeopardise the protection's ability to reliably identify LoM occurrences, making the protection useless.

Options one and two are temporary solutions as other system needs (i.e. fault current, frequency and voltage regulation, grid restoration etc.) will emerge with extremely high IBR penetration [29]. Therefore, option three or combining options one and three by adding SIR (i.e. synchronous condenser placement) and FFR services must be examined. Throughout the EU, only the TSOs of all-island of Ireland have developed and are now delivering joint SIR and FFR products. A study [54] showed that curtailment issues can be partially addressed by reducing the inertial floor (option one) and equipping IBR that can participate in the balancing market (option three).

Table 1 summarises options one and two in some renewable dominant power systems to maintain frequency stability above the UFLS and LoM protection limits. In high inertia power systems, frequency is the metric used to measure system security, and the key concern is to prevent the activation of

TABLE 1 An overview of power system requirements to manage frequency stability.

Regions	Contingency reserve	Inertia floor	|RoCoF| revised limits	SIR service	UFLS trigger
Ireland/Northern Ireland	500 MW	23 GW.s	0.5 -> 1 Hz/s	Yes	48.85 Hz
Great Britain	1250 MW	No – constraint to reduce largest loss is utilised	0.125 -> 1 Hz/s	No	48.8 Hz
USA ERCOT	2 largest generation ~2750 MW	100 GW.s (online monitoring)	N/A	Yes	59.3 Hz
Australia[a]	Regional, SA = 350 MW	Regional limits under risk of islanding. SA = 6 GW.s TAS = 3.8 GW.s	Credible separation = 1 Hz/s Noncredible events = 3 Hz/s	No	47.6 Hz
Nordic	2 largest generators	120 GW.s (online monitoring)	In islanding, the island must withstand 2 Hz/s -> 2.5 Hz/s	No	48.85 Hz

[a]SA: denotes South Australia, TAS: is Tasmania region. It is to be noted that while the National Electricity Market (NEM) does not currently have a system RoCoF limit, generators must comply with the National Electricity Rules, which mandate RoCoF of 4 Hz/s for 250 ms for automatic access and 1 Hz/s for 1 s for minimum access standards [48].

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UFLS. In contrast, low inertia grids rely on RoCoF relays on DGs to protect them against LoM, making RoCoF a significant parameter. As a result, a contingency reserve is always available, to offset an unanticipated trip of a large generator or transmission line while maintaining system balance.

3 | ADDRESSING POWER SYSTEM OPERATIONAL CHALLENGES WITH HIGH IBR

Fast frequency response services have been developed in recent years to address the decreasing system inertia and meet the high renewable energy targets. These services utilise various mitigation measures, such as demand response and IBR.

3.1 | International fast frequency response services

Most of the power grids now face lower inertia and more frequent losses. Moreover, future power systems are anticipated to be operated with substantial amounts of intermittent renewable resources, which could cause the frequency to deviate from nominal value more quickly and expose the system to high RoCoF exposure. Consequently, to manage high RoCoF exposure, new faster-acting frequency response products have been introduced. Table 2 compares technical details of some newly defined international FFR services including the UK National Grid [55], Ireland EirGrid and Northern Ireland SONI [31], Australian Energy Market Operator (AEMO) [56], Canada HQT [57], and Alberta Electric System Operator (AESO) [58], Nordic [59], USA ERCOT [47] and Pennsylvania–New Jersey–Maryland (PJM) [60], and New Zealand Electricity Authority [61]. However, there are currently no widely agreed upon FFR characteristics, with activation time and full delivery time varying across services [37].

National Grid proposed four new fast response services to meet the needs of future power systems. First, DR requires modulated frequency response within 2 s and is aimed at maintaining continuous frequency regulation within a nominal value, as demonstrated in Figure 5 [55]. This service is aimed predominantly at BESS assets to deliver a fast response service. Further, dynamic moderation is a fast dynamic service used to assist with a wider range of frequency regulation and keep it within operational limits. The two services require symmetrical high and low frequency responses. These new products react up to 10 times more quickly than older ones to maintain system frequency at 50 Hz in the face of lower inertia and increasing renewable energy sources.

However, DC and SC services are meant to operate after a major frequency variation. The DC service requires different amounts of low and high frequency containment, depending on infeed and outfeed losses at any given time. This service

TABLE 2 Fast frequency response (FFR) service in different international system operators.

Power systems	System operator	Service name	Service type	Deadband [±mHz]	Service trigger	Response time [s]	For duration [s]
UK	National grid ESO	DC	Dynamic	200	Post-fault contingency	0.5	1200
		DR	Dynamic	15	Post-fault regulation	2	Infinite
		DM	Dynamic	100	Post-fault contingency	0.5	1200
		SC	Static/multi-step	Various	Post-fault contingency	1	1800
Ireland and Northern Ireland	EirGrid and SONI	FFR	Dynamic	15–200	Post-fault contingency	0–2	10
			Static/multi-step	200			
Australia	AEMO	FFR	Dynamic	15	Post-fault contingency	0.1–1	6
			Static/five-step	200		1	N/A
Nordic	SO	FFR	Static/single-step	300, 400, 500	Post-fault contingency	1.3, 1, 0.7	5–30
USA	ERCOT	FFR	Static	15	Post-fault contingency	0.25	900
		LR	Static	300	Post-fault contingency	0.5	3600
	PJM	FFR	RegD	N/A	Post-fault recovery	2	Sustained
Canada	HQT	FFR	Dynamic	100	Post-fault contingency	1.5	9
	AESO	FFR	Static	500	Post-fault contingency	0.2	N/A
New Zealand	Electricity authority	FIR	Static	800	Post-fault contingency	1	60
		SIR		750	Post-fault recovery	60	900

Abbreviations: AEMO: is Australian Energy Market Operator. DC, DR, and DM: are dynamic containment, regulation, and moderation. ESO: is Electricity System Operator. LR: is a load regulation service similar to FFR. PJM: is Pennsylvania–New Jersey–Maryland. SC: is static containment. SO: is a Nordic system operator agreement between Energinet, Fingrid, Statnett, Svenska kraftnät and Kraftnät Åland. Both the LR and FFR in ERCOT have to restore within 3 h and 15 min following the sustained duration, respectively. Additionally, when a resource providing FFR is deployed, it shall not recall its capacity until the frequency is greater than 59.98 Hz [32, 47].

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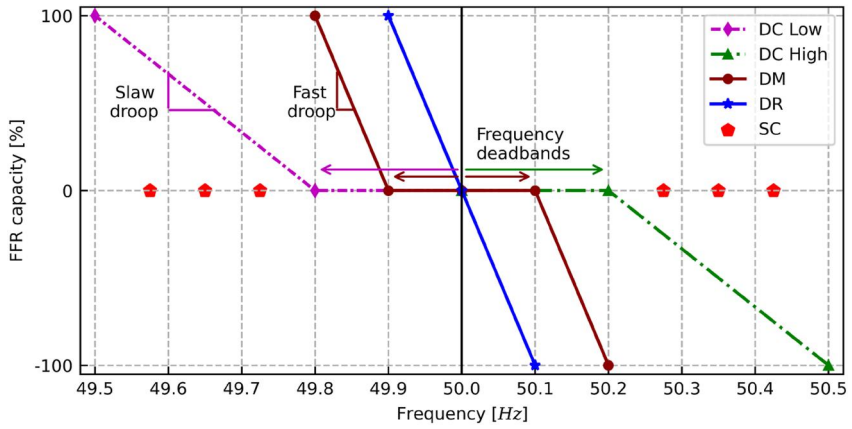


FIGURE 5 Characteristics of dynamic and static fast frequency response (FFR) services in Great Britain (GB).

does not require the symmetrical provision, allowing providers to participate in either low or high frequency alone. It is worth noting that all UK FFR services, except for SC, utilise droop type responses. Droop-based FFR services mimic synchronous generators' response characteristics, with the advantage that converters respond faster than traditional power plants due to smaller delays. [32]. Droop control is a distributed approach that requires no communication between units for coordination, allowing each to contribute in proportion to its size [62].

The UK FFR services respond to both under and over frequency events, unlike the EirGrid FFR service which focuses on limiting the RoCoF and protecting against post-fault under frequency events, as seen in Table 2. The FFR service is defined in the DS3 programme as an additional amount of power delivered or withdrawn from the network in response to response to frequency excursions to improve nadir and RoCoF [31]. One of the technical requirements for this service is that energy delivered between 2 and 10 s must be greater than energy drawn between 10 and 20 s after an incident [63]. IBR, such as wind, solar PV, batteries, and other emerging technologies, including demand response, are suitable for this service.

Nevertheless, the Nordic FFR service prioritises managing maximum instantaneous under frequency disturbances over RoCoF. Static providers in the Nordic region have three activation levels and timeframe options. Both the Nordic static service and LR ERCOT appear to be more suitable for demand response technologies due to their longer post-fault sustainment requirements. Furthermore, the HQT was the first grid operator to require wind farms to integrate FFR features in 2010. It is specifically designed for wind farms with over 10 MW capacity. The system boosts power by 6% above nameplate capacity when activated by high frequency deviation. Further, AESO in Canada developed an FFR service as part of a new technology pilot procurement competition to acquire any technology capable of rapid response. The service initially targeted load shedding and has now been expanded to incorporate BESS.

The TSO of New Zealand also currently procures Fast Instantaneous Reserve (FIR) and Sustained Instantaneous Reserve (SIR) from interruptible loads. The FIR requires interruptible load curtailment within 1 s when the system frequency breaches 49.2 Hz and is required to sustain for at least 60 s following a disturbance. However, the SIR is designed to be delivered within a long time of 60 s following severe disturbances to restore frequency beyond 49.25 Hz. Conventional spinning reserves, it is required to sustain for a minimum of 15 min, whereas, for interruptible loads, they must remain off until instructed by the TSO to provide the service.

3.2 | Frequency response service enabling technologies

Maintaining system frequency security and stability will require a fast response from a wider range of supplementary devices, including renewable technologies, demand response, and energy storage. Keeping the technical characteristics, as discussed in Section 4.1 in mind, the FFR service should be predominantly provided through demand response and IBR. Traditional power plants (e.g., increasing synchronous capacity or adding new interconnectors) are not included in this discussion as they are well-understood. Instead, this section focuses on inverter-based technologies and other sources that may not have an inherent inertial response. The key properties of various resources are summarised in Table 3 [35, 57, 64].

As illustrated, currently, a response time within a range of 250–500 ms from the disturbance time is technically possible from a wide range of devices, and a fast response time is crucial due to lower system inertia resulting in a faster RoCoF. As the FFR service is typically performed via power converters, it can respond quickly, similar to the inertial response of synchronous generators [65]. The FFR service, unlike the inherent response of synchronous machines, is dependent on the active controls and can be tuned to the requirements of the power system within limits. However, the provision of FFR service is limited by the physics and control characteristics of the technology.

To further enhance the response time of supplementary devices in providing FFR service, GFM technologies can be integrated into IBR such as wind, solar PV, energy storage, and HVDC. These technologies can provide fundamental stability enhancements to bulk power systems, including primary

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TABLE 3 Technologies with different frequency response service capabilities.

Technology	Power range (MW)	Energy rating (MWh)	Response time (ms)	Time of sustained power	Maturity
Wind turbines	Depends on the converter if overload and curtailment rates if curtailed	N/A	500–5000	Seconds (through virtual inertia), hours if operated below MPPT	Commercial but limited deployments
Solar PV	Depends on the converter if overload and curtailment rates if curtailed	N/A	100–200	Seconds to hours if operated below MPPT	Relatively immature for controlled active power applications
Flywheel (inverter)	0.1–400	Up to 5	≤4	Up to 15 min	Commercial but limited deployments
Lithium-ion batteries	Up to 50 or more	Up to 20	Few to 40	Minutes to hours	Rapidly moving to mature
Flow batteries	~3, 50 possible	Up to 60	Up to 40	Seconds to hours	Relatively immature
Advanced lead-acid batteries	Up to 40	Up to 40	Few to 40	Seconds to hours	Based on the maturity of technologies
HVDC	Depending on converter capacity	Depending on the contracted period	50–500	Seconds to hours	Mature for triggering response functions
Super-capacitors	Up to 50	Up to 0.28	10–20	Milliseconds to hour	Relatively immature but developing quickly
Demand response	Depending on the contracted volume	N/A	40–60 depends on load type	Seconds to hours (depending on the end user comfort)	Relatively mature technologies for tripping a large number of small loads

frequency regulation, voltage support, and restabilisation of the grid. Grid forming technology allows IBR to set grid voltage and frequency, and even operate without power from the electric grid, making them suitable for providing fast response FFR services. The use of GFM technology in IBR, along with demand response, can play a significant role in providing FFR service and maintaining system frequency security and stability.

3.2.1 | Frequency support from wind power generation

Type 1 fixed speed induction generator (FSIG) wind turbines provide a limited inertial response during disturbances, similar to conventional generators. A FSIG rated over 1 MW typically has an inertia constant of 3–5 s, representing the time it takes for the generator to provide nominal output power from stored kinetic energy [66]. Unlike synchronous generators, VSWT cannot inherently absorb or release kinetic energy stored in their rotors [67], nor respond to frequency regulation signals for system services. Instead, a supplementary control loop is required with the power controller to trigger inertial response and active power reserve during disturbances. The energy stored in the rotating mass can be managed via inertia control techniques [68]. Wind turbine inertial control is classified into two categories: inertia emulation and deloading control technics, as shown in Figure 6.

Emulated inertia

By introducing new control loops, inertia emulation releases kinetic energy stored in rotating masses of wind turbines, temporarily boosting power output. However, the emulated power is limited, lasting 2–6 s [69] which is comparable to a conventional SIR duration of 2–10 s [70]. As kinetic energy is depleted, turbine speed declines, reducing aerodynamic lift and potentially resulting in blade aerodynamic stalling [71]. A recovery period follows, during which output power is curtailed to re-accelerate the turbine and avoid stalling. Two types of inertia emulation techniques exist one-loop, where the controller is triggered by RoCoF measurement, and dual-loops, where the controller responds to both the RoCoF and the frequency divergence [21]. The schematic diagram of both controllers is shown in Figure 7.

The inertial power generated from the one-loop control technique can be mathematically represented using Equation (5).

$$P_{EI} = 2H_{EI}f_{sys}\left(df_{sys}/dt\right) \tag{5}$$

where df_{sys}/dt is the system RoCoF in (Hz/s), H_{EI} depicts emulated inertia in (s) and f_{sys} is the system frequency in (Hz). The inertia control loop is enabled when the controller detects negative or positive RoCoF measurements during frequency deviations.

The dual-control loop differently utilises a combination of RoCoF and system frequency deviation (Δf_{sys}) control loops to

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provide additional power during contingencies. These controllers are mathematically represented as per Equation (6).

$$P_{EI} = K_I\left(df_{sys}/dt\right) + \left(1/K_{droop}\right)\Delta f_{sys} \tag{6}$$

where K_I and K_{droop} are emulated inertia and droop control constant gains, respectively. In this case, the first term of Equation (6) is used to give a short-term emulated power, similar to the one-loop control scheme. This is followed by the activation of the second term in Equation (6), which serves as a droop response to restore system frequency to its nominal value. The latter controller requires the wind turbine to operate at a sub-optimal rate with a specific reserve during pre-disturbance. Notably, the VSWTs can exchange energy with the network faster than synchronous generators due to the quick responsiveness of power electronic devices and a wide range of controller gains. In light of this, a study by Ref. [67] used time varying control gains to regulate the amount of active power injected following frequency deviations, with larger gain values resulting in a better frequency deviation and RoCoF [72].

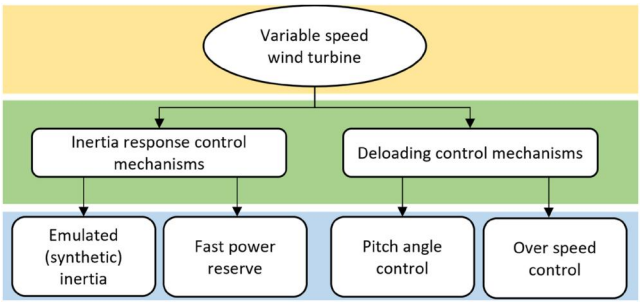


FIGURE 6 Variable speed wind turbine frequency control technics.

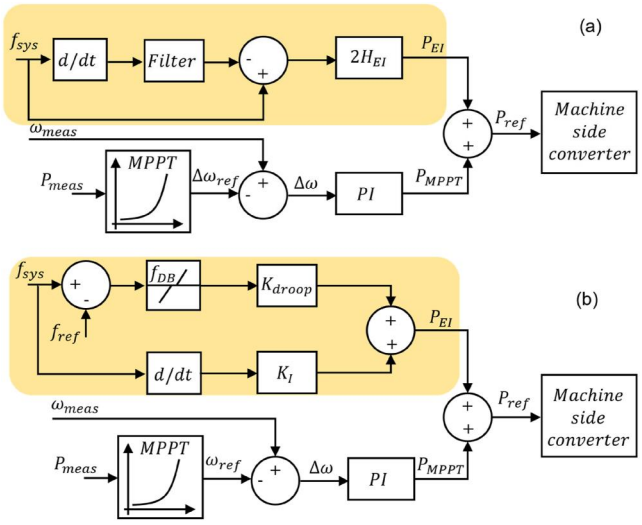


FIGURE 7 Emulated inertia control methods (a) one-loop, (b) dual-loops.

Fast active power

This method utilises a fixed 10% of nominal active power for 10 s, regardless of wind speed, to limit RoCoF and arrest frequency deviations [69]. The rapid power reserve, or short-term constant power, is generated from the kinetic energy stored in the rotating mass of the wind turbine. The active power reserve is obtained by adjusting the set point of the rotor speed, as per Equation (7).

$$P_{const,t} = 1/2\left(j\omega_{r,o}^2\right) - 1/2\left(j\omega_{r,t}^2\right) \tag{7}$$

where $P_{const,t}$ blast active power injected, t ($t < t_{\max}$) is the duration of the sustained power reserve since the frequency event began, $\omega_{r,o}$ and $\omega_{r,t}$ denote the initial and rotor speed at time t, respectively.

The controller is activated when the system frequency deviation exceeds a predefined threshold. The triggering scheme sends a control signal to bypass the MPPT and activate power shaping, as shown in Figure 8 [73]. Additional power is injected, known as overproduction, and when the kinetic energy release is complete, the speed recovery function restores maximum power and returns the rotor speed to its pre-disturbance value. This process frequently results in the underproduction period until rotor speed restores its nominal value. The emulated power and energy recovery depend on wind speed during disturbances [71]. To prevent a second frequency dip, other sources (i.e. BESS and demand response) may be needed to coordinate with the wind turbine response [74, 75]. The transition from overproduction to underproduction must be done gradually to avoid sharp active power declines.

Deloading based droop response techniques

One alternative to inertia response is deloading-based droop response techniques, where the wind turbine is operated below the optimal level to reserve headroom for system service

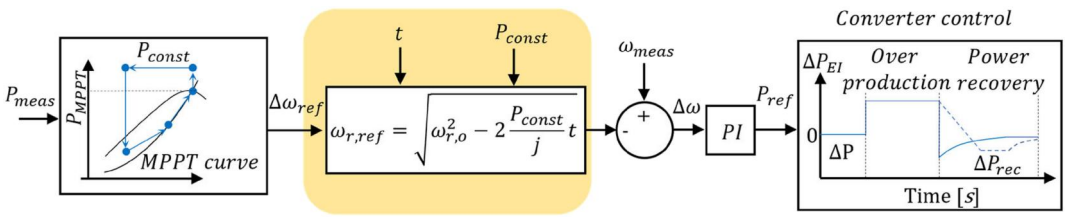


FIGURE 8 Fast active power reserve control diagram and active power characteristics.

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markets during contingencies [76]. This strategy comes at a high energy cost and requires shifting the operating point of the turbine from MPPT to a reduced power level through over-speeding or pitch angle adjustment [77], as illustrated in Figure 9.

The speed control mechanism increases the turbine speed above the MPPT to regulate output power in response to system frequency [73], moving the provided power P_{del} to the right side of the maximum power P_{MPPT}. The rotor speed is then lowered from ω_{del} to ω_{MPPT} to release kinetic energy and inject additional power ΔP. The wind turbine provides a fixed 10% of active power [76], corresponding to a frequency that deviates from the nominal value, with the new operating point between β_1 and β_2 using the reference power P_{ref}, as per Equation (8).

$$P_{ref} = P_{del} + (P_{MPPT} - P_{del}) \frac{\omega_{del} - \omega_{ref}}{\omega_{del} - \omega_{MPPT}} \quad (8)$$

Pitch angle control adjusts the pitch angle of wind turbines to regulate active power generation while maintaining a constant rotor speed to compensate for frequency deviations [78]. It is activated when the wind turbine generator reaches its rated speed and the over-speed controller fails. The pitch angle is increased, moving the operational point from A to a location between A and B. The deloading control technique operates in two scenarios [79]: in steady-state, the VSWT operates at the optimal power curve, while the deloading mode starts as soon as it is switched on. The speed and pitch angle controllers collaborate to store a certain amount of power in various modes.

The droop control mechanism can be used to manage the output power of the wind turbine in proportion to a system frequency deviation [80]. The droop control techniques considerably impact the frequency nadir and quasi-steady-state frequency. The active power generation is adjusted based on droop gain (K_{droop}) using Equation (9). The droop technic is schematically shown in the upper part of Figure 7b.

$$\Delta P_{out} = \frac{1}{K_{droop}} \left(f_{sys} - f_{ref} \right) \quad (9)$$

Wind turbines have been used to provide frequency support in various ways. For example, the Horns Rev 2 Wind Farm in Denmark has been providing primary frequency control to the Danish grid since 2014 using advanced control systems [81]. Additionally, General Electric and the National Renewable Energy Laboratory demonstrated the effectiveness of type-3 wind turbine technology in GFM mode [82]. In January 2019, a two-week trial was conducted in Denmark using a privately-owned wind park consisting of six turbines operating in GFM mode. Another extensive trial took place in May and June 2019 at the Dersalloch wind park in Scotland, which has 23 wind turbines with a maximum power output of 69 MW [83].

3.2.2 | Frequency support from solar photovoltaic systems

Global solar PV capacity is expected to increase by approximately six-fold over the next 10 years, from 480 GW in 2018–2840 GW by 2030, and to 8519 GW by 2050, or almost eighteen-fold 2018 levels [84]. Asia, particularly China leads the market, accounting for more than half of all installed capacity, followed by North America at 20% and the EU at 10%. Most solar PV plants are expected to be integrated into the distribution system, potentially reducing the need for conventional power plants currently used to control system frequency. Solar PV plants are designed to operate at MPPT, which means they typically do not operate with any reserve power currently. However, PV plants are connected to a power system via power electronics converters and can contribute to

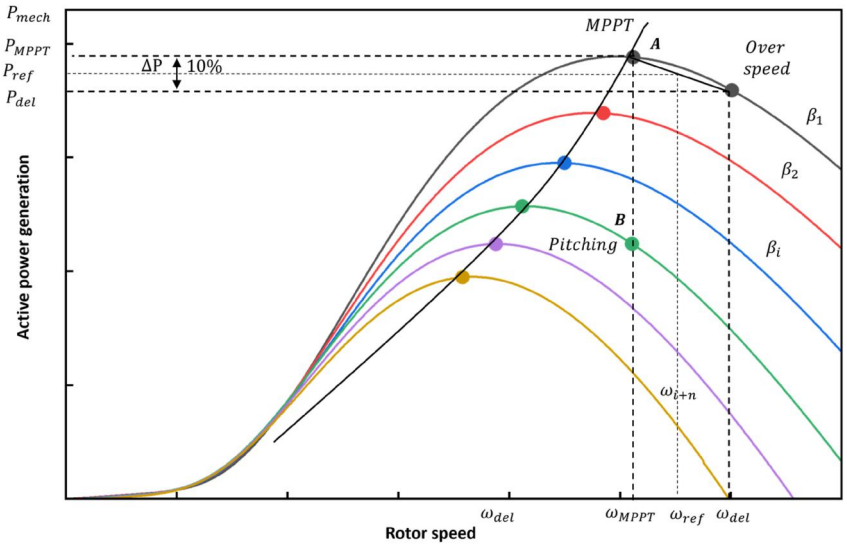


FIGURE 9 Wind turbine deloading techniques include pitch control and overspeeding control.

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system frequency control through the following control techniques [20]:

(1) Solar PV deloading (curtailing plant for reserve).
(2) DC link capacitor response (virtual inertia response).

The most promising control method of frequency management for solar PV facilities is the deloading technique, which is accomplished by raising the PV voltage above the MPPT value. The PV array can curtail some reserve power by boosting the voltage from V_{MPPT} to voltage V_{del}, as illustrated in Figure 10a. This means that the PV plant will be operated at a deloaded operational point P_{del} which is below P_{MMPT}. In such an operational scenario, the curtailed power $\Delta P = P_{MMPT} - P_{del}$ will not be released until the system frequency deviates from the nominal value, as per Figure 10b.

De-loaded operation from the right side region is preferred over the left side region due to improved converter efficiency and faster dynamic response [85]. This is due to the converter's improved efficiency at higher voltage levels and the faster dynamic response brought on by the P-V curve's steeper slope. Furthermore, the left side region is less favourable because of its smaller power margin and instability challenges. However, control systems regulating the DC-link voltage typically use the left side area for PV generation [86]. The left, middle, and right side regions correspond to a current source, power source, and voltage source operation modes, respectively.

The fact that the solid line controller in Figure 10b does not consider the individual PV plant's available reserve is a significant weakness in its design. The fundamental drawback is that, regardless of the reserve power of each plant, all of them will deliver the same quantities of active power required for frequency stabilisation. Consequently, some PV plants with low reserves reach the MPPT sooner and will not be able to contribute to frequency regulation further. To address this issue, a new controller design reflecting the remaining reserve power $V_{reserve}$ has been proposed [87], as shown by the dashed lines. Additionally, an adaptive deloading technique with three controller loops (droop, active power-voltage matching, and vector) has been suggested to quickly regulate the frequency by adjusting the PV system's output power [88].

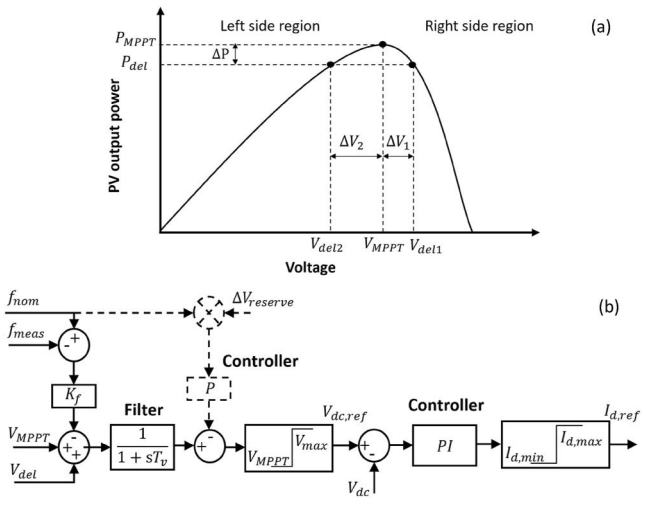


FIGURE 10 Solar photovoltaic (PV) characteristics (a) deloading operational regions (b) active power control system.

Solar PV curtailment can occur when the output of a PV plant exceeds the capacity of the grid or the demand for electricity in the local area. In these cases, the excess energy produced by the PV plant is curtailed or reduced to maintain grid stability and prevent power outages. While solar PV curtailment is generally not desirable, it can be managed in a way that provides value to the grid. For example, curtailed energy can be stored in energy storage systems or used to power other grid services such as electric vehicle charging or water pumping. Curtailment can also assist in correcting over frequency transients similar to the incident that occurred in Queensland, Australia in 2018 [89].

Photovoltaic plants typically do not provide grid regulation directly, but some can provide ancillary services when paired with energy storage or other technologies. One example is the 141 MW solar plant in Copiapó, Chile [90]. While this plant does not have an energy storage system, it has been designed to operate with some flexibility to support grid stability. Another example is the 2.2 GW Huanghe Hydropower Hainan Solar Park in Qinghai, China, which is integrated with 202.8 MW of energy storage capacity for grid services [91]. These examples demonstrate the potential for PV plants to offer ancillary services to the grid and support a reliable and resilient electricity system.

3.2.3 | High voltage direct current interconnectors

In comparison to traditional alternating current (AC) systems, HVDC transmission systems offer improved controllability and are a significant driver for bulk power transmission over long distances. They can be categorised as asynchronous interconnections, embedded transmission lines, and offshore to onshore grid interconnections [92]. Asynchronous HVDC systems are used to link AC networks operating at various frequencies or grids with different phase angles, while embedded systems are used to connect two locations within a synchronous area. High voltage direct current links are also ideal for transporting bulk power from offshore wind farms that are far away such as the BorWin1 in Germany. Examples of these systems include the NORNED HVDC link and the ALEGrO link [93].

VSC-based HVDC converters allow for independent control of active and reactive power, can restore AC electricity after a blackout, and support reactive power similar to static synchronous compensator (STATCOM) devices [94]. They are more dynamic in power reversal and typically smaller in size due to smaller filters. For multi-terminal HVDC grids, VSC-based technology is preferred, and most offshore wind farms use VSC-based HVDC configuration [95]. Offshore converter stations regulate AC voltage and frequency in islanded mode, and voltage source converter (VSC) converters can be used as a current source in the GFL mode [96]. The development of modular multilevel converter (MMC) technology leads VSC

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topologies due to high efficiency, superior output waveforms, and scalability [97]. Modular multilevel converter technology is mostly used in HVDC integration for offshore wind power and can be connected in a GFM mode [98]. However, there are still issues with MMC technology, including high power losses and costly switching equipment.

In HVDC systems, the DC voltage reflects the active power balancing, which can be managed by adjusting the voltage at the station, similar to solar PV. Compared to AC systems, DC systems can react to disturbances far more quickly, in the order of 20 ms, due to the small amount of energy in their capacitors [92]. High voltage direct current systems are also used to assist power system operation, with guidelines developed by ENTSO-E [99]. For example, the all-Island of Ireland power system has two HVDC interconnectors with a capacity of 500 MW each, providing a 75 MW reserve to protect against low and high-frequency response disturbances [51]. The East-West Interconnector, for instance, responds with a rate of 500 MW/Hz when the system frequency drops below 49.8 Hz or reaches 50.2 Hz, up to the maximum agreed-on volume at 49.65 Hz or 50.35 Hz.

3.2.4 | Frequency support from asynchronous and synchronous flywheels

The flywheel is a mechanical device that stores kinetic energy in spinning mass, which is proportional to the square of the rotational speed, as per (10).

$$E = \tfrac{1}{2}\, J\omega^2 \tag{10}$$

where J is the moment of inertia in (kg.m^2) and ω is the angular velocity in (rad/s). Flywheels use grid power to spin a motor at high speeds and store energy. To release energy, the momentum of the spinning flywheel forces the motor to behave as a generator, slowing down the spinning and injecting power into the grid [100]. This process provides inertia support. The amount of kinetic energy stored depends on the physical attributes of the flywheel, including its speed and inertia.

An asynchronous flywheel can provide short-term energy storage and is ideal for continuous charging and discharging processes, such as frequency regulation, to support the integration of large-scale IBR [101]. Flywheels are well-suited for applications that require near-constant cycling to maintain a balanced power system, addressing short-term energy balancing and power quality issues. Flywheels can provide advantages over BESS and super-capacitors in terms of complete depth of discharge cycles (approximately 200,000 vs. 5000 for BESS) and energy density, respectively [14, 15]. Flywheels also offer fast-ramping capabilities, high roundtrip efficiency, and a lifespan of 20 years or more. However, flywheels have limitations in terms of energy storage capacity.

Synchronous condensers offer a different approach to energy storage and grid stabilisation compared to flywheels. They are unloaded synchronous machines that operate in the motoring mode by default but can switch to the generating mode instantly if the frequency falls Despite the emergence of new technologies such as static var compensators that offer better controllability and lower cost, synchronous condensers continue to be used in many areas worldwide to enable the growth of renewable energy [102].

The market for synchronous condensers is expected to grow from its current value of US$ 1.01 Billion in 2022 to US$ 1.76 Billion in 2029 [103]. This growth is mainly driven by the integration of renewable IBR and the use of synchronous condensers to support the reliable operation of power plants during the decommissioning process. While IBR are capable of replacing ancillary system services that traditional power plants provide, there are still some functionalities of the synchronous condenser that IBR cannot replace. Table 4 provides a comparison of the capabilities of both technologies [29, 34, 38].

The decommissioning of mostly coal-based power stations has led to low short circuit levels and reduced inertia, which has driven a resurgence in interest in synchronous condensers. Redundant synchronous generators from these power stations can be converted into synchronous condensers with inertia augmentation through flywheel installations. Synchronous condensers add inertia to the system by reacting quickly to power imbalances and contributing to RoCoF's constraints

TABLE 4 Technical capabilities of synchronous condensers versus IBR.

Synchronous condenser	Inverter based resources
Synchronous condensers have voltage source characteristics, which means they can maintain the terminal voltage	Have physical limitations in terms of current and voltage control capabilities
Have a considerable overloading capability and short-circuit current injection	Only provide a small overloaded current rate for a short timeframe
Have inherent inertia support functions	Can support emulated inertia
Standard synchronous machine control mechanisms	The controller has a high bandwidth, can provide fast frequency and voltage regulation, and can perform numerous control schemes
They are physically limited and can only absorb a part of the full rated current	Can deliver the full rated current in both inductive and capacitive modes
Does not interact with other control systems	Issues with converter stability can arise from power converters' quick response time interacting with controllers of the same type and other networked devices

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[104], and they provide a variety of other services to tackle issues, including frequency control, transient stability, supporting grid voltage, and short circuit strength. They also enable power system protection, reactive power support for dynamic voltage control with robust overloading and fault ride-through capability [37], and oscillation damping at the short circuit level [105]. These are the key reasons why this technology is being "resurrected" and attracting widespread interest in islanded power systems (i.e. in Ireland, the UK, Australia, and Denmark), as shown in Table 5. The services provided by synchronous condensers can help make up for the loss of inertia and other services provided by the decommissioned power plants, ensuring that the system has enough strength to operate steadily.

It is worth noting that flywheels alone cannot alleviate frequency stability challenges, but they can be significant when combined with other mitigation measures such as STATCOM and BESS, as seen in the Phoenix project in Scotland. The features of the two systems can be combined to create a new technology that can do a variety of tasks and offer a range of services. However, such a system would not contribute as much in terms of overloading capability as a standalone synchronous condenser with a similar rating and coordinating the control of the combined system would be more challenging.

3.2.5 | Battery energy storage systems

Energy storage is widely recognised as a way to enhance grid stability, especially if featured with pre-designed control requirements that enable participation in new services such as FFR and inertia response. A grid-scale BESS can provide both frequency and voltage support to the network following disturbances, as well as act as an energy balancing resource. One key advantage of BESS is its ability to work in four quadrants, meaning it can consume or generate active and reactive power, providing fast response times (usually between 100–150 ms or even less) during system contingencies, and delivering grid services with a high degree of flexibility [113]. For instance, an 11 MW/5.5 MWh BESS installed in County Kerry, Ireland, can provide FFR and both active and dynamic reactive power services with a response time of less than 150 ms [114]. Similarly, a 10 MW/5 MWh lithium-ion battery deployed by AES at Kilroot Power Station in Northern Ireland in 2016, primarily as a prototype plan to respond to system frequency disturbances, effectively providing both frequency regulation [115]. This deployment has been shown to offer ancillary services for grid stability during disturbances, as shown in Figure 11, which corresponds to two real frequency events captured by phasor measurement units.

TABLE 5 Recent synchronous condenser projects in islanded power systems.

Country	Project name	Commission date	Installed capacity	Objective	Reference
Ireland	Green Atlantic at Moneypoint by Siemens and electricity supply board (ESB)	2022	400 MVA including large flywheel weight 130 tonne	Allow for higher penetration of wind power into the Irish grid	[106]
UK	Rassau project in Wales by Siemens Energy and Quinbrook	2024	Combined 100 MVA synchronous condenser and 239 tonnes flywheel	In response to the rapid increase of renewable energy sources, the project is intended to play a significant role in National Grid's innovative approach to power system stability while attaining the net zero target	[107]
	Lister driver project in Liverpool by ABBand Statkraft	2021	Combined 67 MVAr synchronous condensers with 40 tonnes of a flywheel	Will provide more than 900 MWs of inertia and increase the instantaneously available inertia by 3.5 times	[108]
	Phoenix project in Neilston by ABB project partner	2022	Combined 70 MVA voltage source converter (VSC) STATCOM with 70 MVA synchronous condenser	Referred to Hybrid-synchronous compensator (H-SC), can also provide reactive power support for black start	[109]
Australia	Davenport project manufactured by GE	2021	Two synchronous condensers rating 575 MVA/plant	Each plant provides 1100 MW.s and in total, they provide 4.4 GW.s of inertia. To maintain RoCoF below 3 Hz/s, secure operating envelope for voltage and transient stability	[110]
	Robertstown project	2021	Two synchronous condensers rating 575 MVA/plant		
	EnergyConnect, interconnector by ANDRITZ	2024	Four synchronous condensers rating 120 MVA/plant	Inertia contribution is 7 MW.s/MVA	[111]
Denmark	Suðuroy, Faroe Islands project by ABB	2022	Combined 7.5 MW BESS and 8 MVA synchronous condenser	Could enable 100% of the island's demand to be met by wind energy	[112]
	Sund, Faroe Islands	2023	Synchronous condenser		

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Figure 11 provides a clear demonstration of the BESS's ability to respond quickly to frequency disturbances, as seen in the top panel, where the BESS absorbed active power soon after the system frequency exceeded the controller deadband limit of 50.15 Hz. The BESS continued to consume power from the network until the system frequency returned to below the controller sensitivity band. The bottom panel shows the BESS's response to a cascaded under-frequency event, with the first trip occurring initially outside the BESS deadband limit, resulting in a delayed response, whereas the second trip, within the deadband limit, resulting in a faster response. Notably, due to the lack of system reserve, the frequency recovered to nominal after an extremely long time during this event.

Small-scale BESS devices, such as those with a capacity of 10 kW, can be deployed to reduce network restrictions and avoid costly network upgrades, which can negatively impact revenue streams and make it challenging to determine suitable installation locations [113]. However, the next phase of storage development involves deploying hundreds of MWs of BESS at single sites. The Hornsdale power reserve in South Australia is well known example of this, featuring a 100 MW/130 MWh Li-ion battery integrated with PV systems. The reserve effectively provides system services during low-inertia power dispatches, with 70% of its capacity reserved for providing FFR services to Australia's NEM, while the remaining capacity participates in the conventional energy market [116].

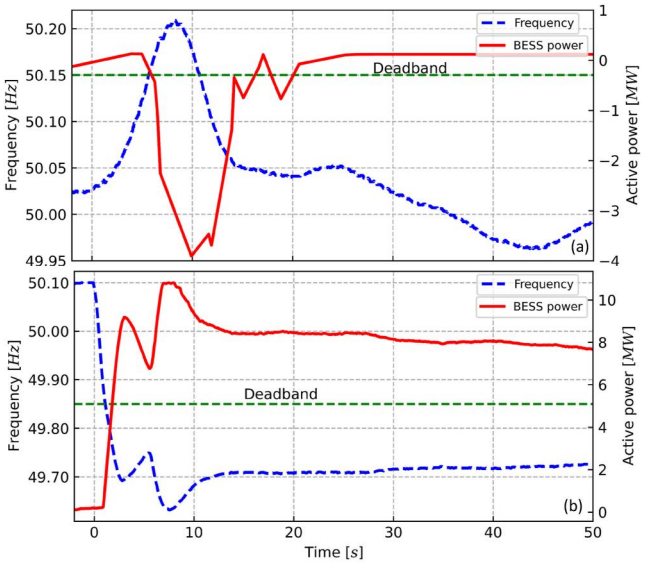


FIGURE 11 Example of battery energy storage systems (BESS) active power response to frequency disturbances at Kilroot power station, (a) over-frequency occurred on 04/03/2016, (b) cascaded under-frequency occurred on 22/11/2021.

In power systems, IBR such as solar PV, wind power, HVDC, and BESS can use either GFM or GFL converter technologies. A GFM converter allows independent direct control of terminal voltage from grid voltage [38], unlike GFL converters where the current is directly regulated. In contrast to GFL technology, which requires a network signal for synchronisation and current injection, GFM technology can generate a voltage signal even in the absence of a load or grid voltage. These features make the GFM technology well-suited for enhancing the frequency response based on the dynamic capability of the converter, as described in Table 6 [34, 117–119].

In the literature, various GFM solutions have been proposed and implemented, such as virtual synchronous machines (VSM) [120], virtual oscillator based control [121], power synchronisation control [122], synchronverters, direct power control, and matching control [123]. However, some basic control approaches involve allowing energy storage units to produce proportional $p - f$ droop responses [38]. Although the GFM converters have been widely utilised in microgrids, standalone systems, and uninterruptable power supply systems [124], their application has recently extended to energy storage and IBR [125]. For instance, the Dalrymple 30 MW/8 MWh BESS is the first grid-connected battery-based GFM converter in the NEM, providing both regulated and competitive market services [126]. It is based on VSM technology, which mimics the behaviour and performance of synchronous generators, delivering both synthetic inertia and high fault current to

TABLE 6 Comparison between grid-following mode (GFL) and grid forming (GFM) grid support capability.

Grid following capability	Grid forming capability
Act as a current source and under typical operations, it is assumed that the grid has already been established.	Act as a voltage source while forming and maintaining a robust grid (e.g., contributing to system strength)
Require PLL to synchronize with the network	It can operate without PLL as well as with PLL to switch between different control modes
Follow network voltage to provide active and reactive power	Can provide the black-start capability from a complete shutdown
Control of current delivered to the grid using a direct-quadrature (dq[a]) vector	Able to control voltage magnitude and frequency/phase using $P - f$, $Q - V$
Cannot actively operate at 100% IBR power penetration	Can hypothetically function with 100% IBR as well as cooperatively operate with grid following systems
Well established and globally standardized	Lacking operating experience from a systems operator point of view, as it is not standardized.
Commercially are not available with high overcurrent capability (even though technologically possible)	Commercially available to provide fault current of 200% of the rated value at the inverter terminals

[a]dq control implies control concerning a three-phase synchronous reference frame. $P - f$ refers to active power and frequency. $Q - V$ reactive power and voltage.

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accommodate more renewables into the network [127]. More pilot projects and demonstrators are being established worldwide, as summarised in Table 7 [128].

3.3 | Frequency response activation techniques

The power system frequency response services are typically classified into two parts based on the activation method (e.g. droop and direct static step changes in the reference power), as shown in Figure 12.

The fundamental principle of droop control is to raise the active power reference when the grid frequency is lower than the nominal value and lower the active power reference when the grid frequency is higher than the nominal value [27]. This can have linear features, as illustrated in Figure 13a [129]. The predetermined droop coefficient, frequency limit, power limit, and active power reference are used to regulate the active power reference. Droop parameters between 3% and 5% and a deadband of not more than 17–36 mHz are often applied [64].

Pre-curtailment is an important issue for IBR providing upward reserve services from renewable energy, resulting in lost energy sales. To alleviate this issue, droop coefficients can be treated individually in the over and under-frequency domains, as illustrated in Figure 13b. A deadband was added to prevent frequent activation due to measurement tolerance or sensor noise, as shown in Figure 13c [130]. However, fixed droop rates can be further modified using the pointwise approach, where the droop curve is created by connecting the dots, allowing for the expression of p-f linear characteristics [131]. Alternatively, segmented droop control can be used, where smaller droop coefficients restore minor frequency deviations, and larger droop coefficients are used for extreme deviations, as illustrated in Figure 13d.

Nonetheless, the standard droop may not be suitable for providing FFR service as they are specially designed for slower response POR. From the perspective of TSO, a more aggressive droop with a narrower deadband can quickly restore grid frequency [132]. The frequency magnitude [133] or RoCoF [134] are commonly used to trigger FFR services based on a static step response as in Figure 13e. However, abrupt changes in active power can cause system oscillatory behaviour,

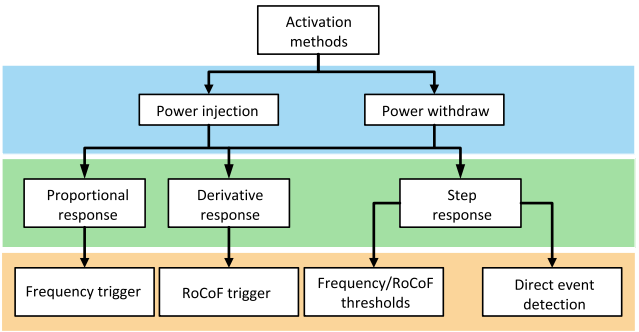


FIGURE 12 Methods deployed to activate frequency response services.

TABLE 7 Global pilot grid forming (GFM) projects.

Country	Project	Capacity	IBR type	Market type
UK	Dersalloch Siemens-Gamesa project in Scotland	69 MW in VSM mode connected to 33 kV substation	Wind farm	Ancillary service, black start and islanded mode
Australia	Hornsdale power reserve, Jamestown	150 MW/194 MWh in VSM mode connected to 275 kV substation	Lithium-ion BESS	Ancillary service and energy market
	Energy Storage for Commercial Renewable Integration (ESCRI) South Australia project by Hitachi ABB, near Dalrymple	30 MW/8 MWh in VSM mode connected to 132 kV substation	Lithium-ion BESS	Ancillary service and energy market
Switzerland	OSMOSE project, Ecole Polythecnique Fédérale de Lausanne	720 kW/560 kWh, connected to MV 0.3 kV substation	Lithium-Titanate Oxide BESS	Renewable integration
	OSMOSE project, Ecole Polytechnique Fédérale de Lausanne (EPFL) campus	25 kWh, connected to 400 V microgrid	Lithium-Titanate Oxide BESS	Renewable integration
France	OSMOSE project, RingoLab	6 x 0.5 MVA/60 min BESS and 6 x 1 MW/10 s supercapacitors	Hybrid energy storage system (HESS)	Renewable integration
Spain	La Plana Siemens-Gamesa project	3 x 435 kW/145 kWh lithium-ion BESS and 1 x 120 kW/400 kWh Redox-flow	Hybrid	Ancillary service, black start and islanded mode
Portugal	DEMOCRAT project	2 x 200 kW/109 kWh, connected to LV 400 V and MV 15 kV	Lithium-ion BESS	Black start, energy market and islanded mode
USA	SMA st. Eustatius project	5.9 MWh	Lithium-ion BESS	Renewable integration and islanded mode
	NREL campus Colorado project	1.25 MW/1.25 MWh and 1 MW/1 MWh connected to 13.2 kV	BESS	Ancillary service and renewable integration

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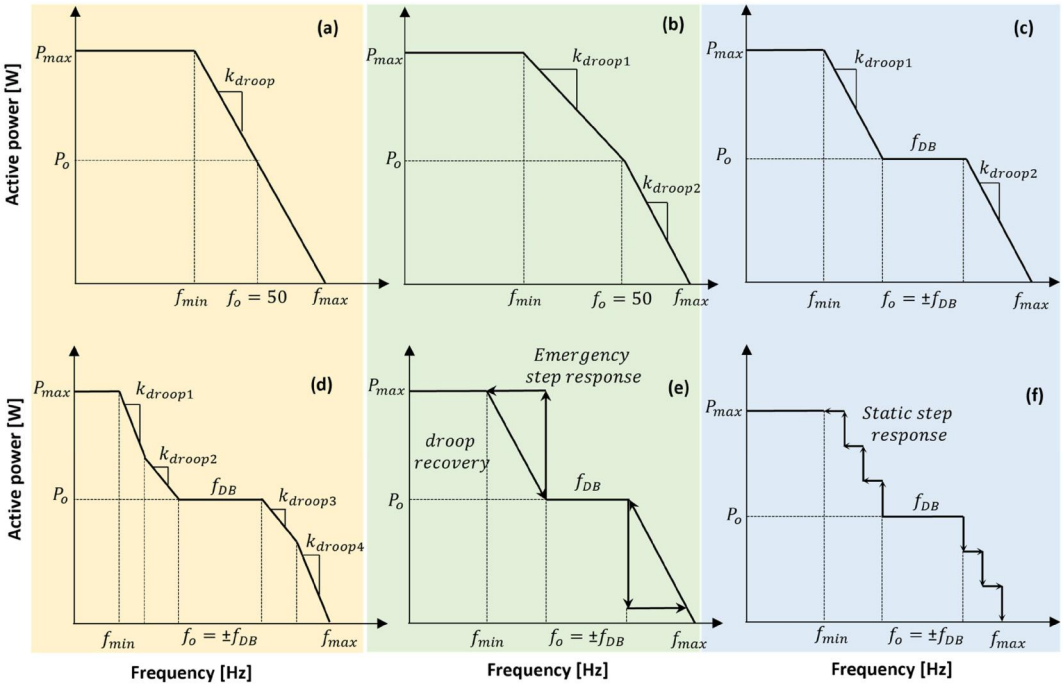


FIGURE 13 Frequency droop control methods.

so the power is progressively restored in multiple steps [14, 15], as shown in Figure 13f. In emergencies, a centralised activation signal can cause a portion of demand to trip in a single step, but the active power must be restored gradually to avoid a second frequency disturbance. In Ref. [135], Two-stage emergency control techniques on RoCoF constraints were used to arrest frequency deviation.

Compared to droop controllers, this static response has the advantage of the fastest reaction to the specified power quantity once activated. It has the potential to reduce spinning reserve requirements under low inertia dispatches by quickly balancing power through the step response of some sources [136]. However, this control can only assist with extreme frequency deviations and cannot continuously control the frequency to the desired level, unlike droop-based control [14, 15]. Coordination among all participating resources may be essential to achieve the best performance [137]. Additionally, the static step response takes some time to reset and re-arm once activated, whereas a droop-based control system will respond as long as there is headroom and energy available.

Using frequency signals to activate demand response is more accurate than utilising RoCoF because RoCoF is the second derivative of phase angle (e.g. voltage or current) and can be noisy, requiring sample interval filtering [138, 139]. Although there are various techniques available with different accuracy and performance trade-offs (e.g. the interval between zero crossings and fast Fourier transforms), there are always concerns about false triggering based on RoCoF [21, 140]. Furthermore, as power systems' inertia decreases, even minor disruptions can cause extreme nadir and rapid RoCoF [134]. This can result in excessive wear and tear on over-frequency as a result of tripping demand greater than the required capacity. Therefore, during ultrahigh RoCoF levels caused by major disturbances, the FFR service triggered through system frequency measurement is unlikely to achieve the desired outcome [35].

As a result, FFR for low inertia conditions is expected to be triggered mainly by direct event detection rather than local frequency or RoCoF detection [35]. This can be achieved through communication response triggers or wide-area broadcast options [141, 142]. Both approaches require pre-programed action to inject arresting energy into the grid, which bypasses the problems with measuring frequency and/or RoCoF. However, they come with their own sets of challenges such as relying on grid topologies, communication system performance, and security. Some service providers may still use frequency measurements to trigger FFR to preserve system security.

3.4 | Distributed generation in IEEE standards

The typical view of electricity infrastructures is of grid-scale central power plants capable of generating one direction electricity flow through high-voltage transmission lines over long distances to distribution networks, which are then distributed to a large number of consumers. Due to the introduction and advancement of DG based IBR, the traditional system structure has been altered, resulting in two-way power flow due to dispersed IBR throughout the network. Consequently, several organisations have established technical standards for integrating IBR interconnections to ensure system stability, as depicted in Figure 14. For example, the IEEE Standards 1547 series has served as a fundamental document for integrating

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IBR into the power system. The first series IEEE 1547-2003 primarily specifies the minimum technical requirements, such as power quality, frequency deviations, and voltage levels as well as safety and protection, that are universally required to ensure secure and reliable connectivity [143].

Due to the potential impact of the increased penetration of IBR on the power system operation and dynamics, standard 2013 presents an overview of distribution secondary network system design, discusses considerations for connecting IBR to networks, and suggests potential solutions for IBR connectivity in network distribution systems [144]. The objective of standard 2014 was to manage the potential impact of IBR operation at the system level. Nonetheless, this project was withdrawn later in 2016. The interconnection standards in place were adequate for small amounts of power flows from IBR but not for the large number of IBR anticipated to support the fundamental change within the distribution system. IEEE standards and grid codes worldwide have been fundamentally changing to require IBR to provide power system support services and remain connected for a particular time when a grid malfunction occurs to boost the hosting capacity of renewable IBR.

This has resulted in a series of amendments (i.e. IEEE 1547a-2014 and IEEE 1547.1a-2015) to support low/high voltage/frequency ride-through functions (L/HVRT and L/HFRT) in response to faults at the terminal of IBR units [145, 146]. To handle compatibility and associated interfaces between IBR and the network, standard IEEE 1547a-2014 demands the power system support functions, including frequency-droop regulation and inertial response to the RoCoF. In IEEE 1547-2018, if reserve power is available, the IBR must provide frequency-Watt (p-f) control, but it is not required to maintain an upward or downward reserve to enable $p - f$ [147].

Furthermore, IEEE 2030 is a foundational set of system-of-systems standards developed from scratch in 2011 to govern network connectivity and interoperability [148]. It has been modified to allow for the continued implementation of

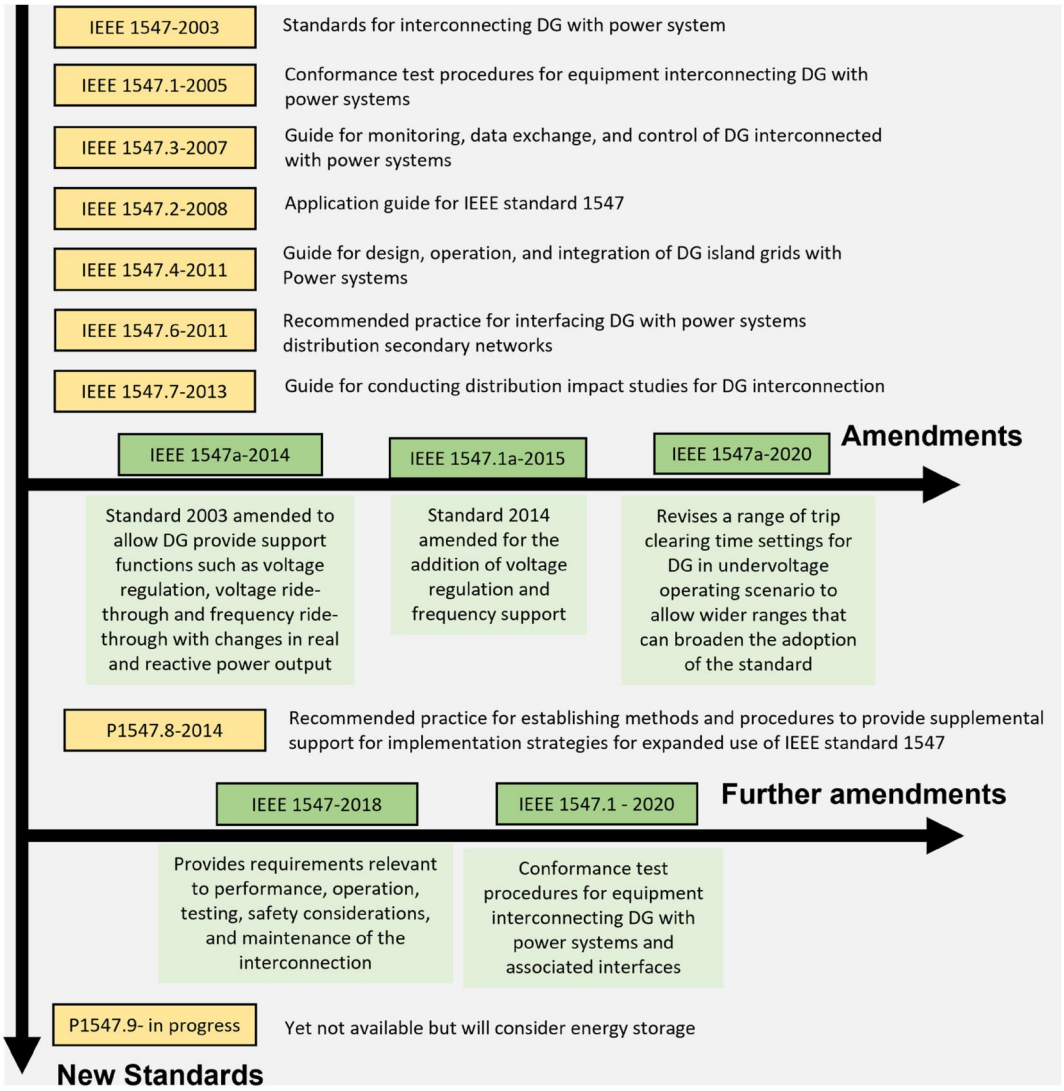


FIGURE 14 The IEEE standard developments for managing distributed generation (DG) operations.

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modern communications and information technologies that provide interoperability solutions for IBR connectivity as well as smart end user appliances. Detailed guidance is provided in series 2030-1 up to 2030.3 for power system support functions by electric transportation infrastructure and interoperability of energy storage systems. The revisions primarily concern the ability of two or more systems to exchange and use information securely and effectively. These standards have addressed the participation of IBR technologies including the potential grid-scale data centre in the future power system operation with high penetration of renewable IBR. Finally, a clear path has yet to be identified for IBR equipped with different converter technologies (i.e. GFL and GFM), including their share and allocations to securely operate renewable rich power systems.

4 | DISCUSSIONS AND RECOMMENDATIONS

The ongoing transition towards 100% renewable electricity generation presents both opportunities and challenges for grid operators. The positive aspect of renewable energy resources is the potential to reduce greenhouse gas emissions and improve the sustainability of the power grid. The negative aspect is the increasing share of IBR, combined with the progressive decline in system inertia, poses significant concerns for system stability and reliability. In particular, maintaining frequency deviations and RoCoF within acceptable limits has become increasingly difficult. The traditional provision of frequency control services through synchronous generators is becoming inadequate and expensive, necessitating new approaches to ensure the stability of the power grid. To address these challenges, grid operators can take some steps to securely approach 100% renewable electricity generation.

- First, investing in energy storage technologies can help provide critical grid services such as frequency regulation, voltage support, and ramping control. By encouraging the deployment of energy storage technologies, grid operators can improve the reliability and resilience of the power grid.
- Second, consider the deployment of synchronous compensators to provide grid support services. Synchronous compensators can help to address the challenges associated with the decline in system inertia by providing additional inertia support to the grid. They can also help to mitigate voltage fluctuations and provide reactive power compensation, which is essential for maintaining voltage stability in the power system.
- Another key strategy is the implementation of advanced control systems that can manage the variability of renewable energy resources and ensure system stability. These control systems can forecast renewable energy generation and demand, manage power flows, and coordinate the operation of distributed energy resources. Additionally, flexible grid infrastructure, such as demand response, dynamic line rating, and grid-forming inverters, can help accommodate changes in demand and supply and improve the stability and reliability of the power grid.
- Fostering collaboration among stakeholders is also crucial for ensuring the secure integration of 100% renewable electricity generation. By establishing joint working groups, sharing data and information, and developing common standards and guidelines, grid operators can work with renewable energy developers, energy storage providers, and other stakeholders to facilitate the exchange of best practices and lessons learnt.
- Finally, developing future control rooms for monitoring and controlling renewable energy resources is essential for ensuring the stability of the power grid. These control rooms will need to be equipped with advanced software and hardware technologies that can effectively manage the complex and dynamic interactions between various renewable energy resources and the power grid. By incorporating advanced analytics tools, machine learning algorithms, and specialised monitoring equipment, system operators can accurately predict and anticipate changes in renewable energy generation and demand, thus proactively adjusting the balance between energy supply and demand.

5 | CONCLUSIONS

This paper provided an extensive overview of power system stability issues as synchronous power plants are rapidly being displaced by IBR. This research evaluated a detailed power system dynamic frequency stability issue associated with integrating IBR and solutions in prospective technological pathways to assist system operators. With more widely acknowledged IBR values and benefits (i.e. improved controllability and grid resilience) global power systems are transitioning to a new energy mix that relies less on fossil fuels and nuclear while incorporating more energy storage, demand response, and IBR. From reviewing the key FFR enabling technologies, it was possible to conclude that power systems require new sources of flexibility to accelerate the transition to high IBR via participation in FFR service markets. These flexibilities can come from IBR with advanced converter control mechanisms or from synchronous condensers. Furthermore, addressing power system scarcities will necessitate a holistic approach based on system-level thinking while introducing new markets incentivising investment in technology that can help the electricity system transition to a cleaner and greener future. Finally, future research should explore the long-term frequency stability implications of high penetration of IBR technologies to ensure the overall stability and reliability of the power system.

AUTHOR CONTRIBUTIONS

Dr Dlzar Al Kez is a research fellow at Queen's University Belfast wrote the manuscript as part of PhD in Power System Dynamics with Increasing DG Penetrations with significant

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inputs and contributions from Prof Aoife M. Foley supervisor. Dlzar Al Kez: investigation; writing – original draft; visualisation. Aoife M Foley: supervision; conceptualisation; investigation; writing – review & editing; funding acquisition. Faraedoon W. Ahmed: reviewing & editing. D. John Morrow: supervision; conceptualisation; investigation; writing – review & editing; funding acquisition.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

No data were used or analysed in this review. This paper is a conceptual review based on analysis and synthesis of existing literature in the field.

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